

The Complete Original Gamma Return

Low endpoint, parity and positive contiguous transport

Split-Zero research programme

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This supplement prints the complete low-endpoint transport, parity and Toda transport, and independent contiguous determinant derivations. The original moments, masses, coefficient maps, four endpoint indices, and parameter domains are retained. Complete prerequisite proof sources and current receiving statements accompany the editable text.

The original low Gamma relation endpoint

This calculation continues the full-source comparison in BRD1–42 and CTR1–54. It treats the original one-column relation determinant separately, so the degree condition $r \geq l$ of the projector construction is never applied to the low endpoint. Every comparison below is on the original scalar coefficient space, with the original polynomial multiplication maps and all four tail integrals present. The complete BRD, FTC and CTR sources supply the proved contour, polynomial and high-endpoint formulas.

1 Objects and the low source maps

Retain

$$k = 4l + 1, \quad l \geq 1, \quad m \geq 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad c = k/2, \quad 0 < \delta < 1/2, \quad \gamma > 2. \quad (\text{LET1})$$

Set $d = \delta^2$, $g = \gamma^2$, and define the complete original objects

$$\begin{aligned} \chi(S) &= \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^e, \\ f(S) &= \prod_{j=0}^{l-1} (S - c - t_j), \quad t_j = 2j + \frac{1}{2}, \quad \sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}. \end{aligned} \quad (\text{LET2})$$

Write $m_j = \int_{\mathbb{R}} y^j \sigma(y) dy$, with the index denoting the literal power of y . Thus $m_0 = \sqrt{2\pi}$, odd moments vanish, and CTR50–53 supply all even moments by their exact rational recurrence with the full mass factor. The original low relation endpoint is

$$\mathfrak{r}_\chi = \frac{\int_{\mathbb{R}} |f(c + iy)\chi(c + iy)|^2 \sigma(y) dy}{\int_{\mathbb{R}} |\chi(c + iy)|^2 \sigma(y) dy}. \quad (\text{LET3})$$

All four scalar integrals used below are finite by the proved Gamma upper bound and strictly positive because their polynomial factors vanish at only finitely many real points.

Put $\epsilon = 2^{-10}$, and partition the line into $I = \{|y| < q\epsilon\}$, $B = \{q\epsilon \leq |y| \leq 64q\}$, and $F = \{|y| > 64q\}$. Assume the explicit finite threshold

$$k \geq 2048\sqrt{d + g}. \quad (\text{LET4})$$

Then $q \geq k^2$ gives $k\sqrt{d+g}/q \leq \epsilon/2$. Also $t_{l-1} < k/2$ and $\sqrt{d+g} > 2$ give $t_{l-1}/q \leq \epsilon$. This is the same original-parameter domain already used for the complete high-endpoint comparison.

On the bulk Hilbert space $L^2(B, \sigma(y) dy)$, the maps from the one-dimensional coefficient space are

$$T_{\text{pow}}u(y) = (iy)^q u, \quad T_\chi u(y) = \chi(c + iy)u, \quad (Uh)(y) = \frac{\chi(c + iy)}{(iy)^q} h(y). \quad (\text{LET5})$$

Their exact identities are $UT_{\text{pow}} = T_\chi$ and $UM_f = M_f U$, where M_f is multiplication by $f(c + iy)$. Both U and its inverse are bounded: the bulk is compact, $y \neq 0$ there, and LET4 places every displacement root strictly inside its inner radius. The squared modulus of U is the original density ratio. Taking scalar Hilbert Grams proves the source comparisons below while retaining the original arithmetic quotient and its multiplication maps.

2 A scalar ratio transport retaining the degree l

Let w be a finite positive measure, $F_0 \geq 0$ a measurable function with $0 < \int F_0 dw < \infty$, and let ψ be a bounded real function. Define the two literal ratios

$$R_w = \frac{\int F_0 dw}{\int dw}, \quad R_{e^\psi w} = \frac{\int F_0 e^\psi dw}{\int e^\psi dw}.$$

If $a \leq \psi \leq b$ almost everywhere, positivity of the integrals gives

$$e^a \leq \frac{\int F_0 e^\psi dw}{\int F_0 dw} \leq e^b, \quad e^a \leq \frac{\int e^\psi dw}{\int dw} \leq e^b.$$

Division and logarithms therefore prove the exact finite estimate

$$|\log R_{e^\psi w} - \log R_w| \leq b - a. \quad (\text{LET6})$$

No restriction on the degree of $F_0 = |f|^2$ enters this scalar proof.

Apply LET6 to $dw = 1_B(y)|y|^{2q}\sigma(y) dy$ and $F_0(y) = |f(c + iy)|^2$. For $y = qz$, the unchanged root product has the expansion

$$\begin{aligned} \psi_\chi(qz) &:= \log \frac{|\chi(c + iqz)|^2}{|qz|^{2q}} = \frac{a_k}{z^2} + E_\chi(qz), \quad a_k = \frac{k(k+2)(d-g)}{3q}, \quad |E_\chi(qz)| \leq \delta_\chi, \\ \tau_\epsilon &= \frac{k^2(d+g)}{q^2\epsilon^2} \leq \frac{1}{4}, \quad \delta_\chi = \frac{q\tau_\epsilon^2}{2(1-\tau_\epsilon)}. \end{aligned} \quad (\text{LET7})$$

For completeness, write each factor as $iqz(1 + i\zeta_{ab}/(qz))$. The logarithmic Taylor series converges absolutely; the permutation $(a, b) \mapsto (k-a, k-b)$ cancels odd powers. The quadratic sum is $e \operatorname{Re} \sum_{a,b} \zeta_{ab}^2 = qk(k+2)(d-g)/3$. The remaining even powers have absolute sum at most $q \sum_{j \geq 2} \tau_\epsilon^j/j \leq q\tau_\epsilon^2/[2(1-\tau_\epsilon)]$. This proves every coefficient and remainder in LET7.

Since $z^{-2} \in [64^{-2}, \epsilon^{-2}]$, define the finite constant

$$E_0 = |a_k|(\epsilon^{-2} - 64^{-2}) + 2\delta_\chi. \quad (\text{LET8})$$

The exact bulk difference

$$d_\chi = \log \frac{\int_B |f\chi|^2 d\sigma}{\int_B |\chi|^2 d\sigma} - \log \frac{\int_B |f(c + iy)|^2 |y|^{2q} d\sigma}{\int_B |y|^{2q} d\sigma}$$

satisfies $|d_\chi| \leq E_0$ by LET6. All products in the first ratio are evaluated at $c + iy$. The original numerator multiplication also gives

$$1 \leq \frac{|f(c + iy)|^2}{|y|^{2l}} = \prod_{j=0}^{l-1} (1 + t_j^2/y^2) \leq e^{\Delta_f}, \quad \Delta_f = \frac{S_2}{q^2\epsilon^2}, \quad S_2 = \sum_{j=0}^{l-1} t_j^2 = \frac{l(16l^2 - 12l - 1)}{12}. \quad (\text{LET9})$$

The upper bound follows by integrating the inequality $(\log(1+x))' = 1/(1+x) \leq 1$ from zero. Expanding each t_j^2 and summing j, j^2 gives the stated S_2 . Consequently the logarithmic correction d_f obtained by replacing the bulk numerator $|f(c+iy)|^2|y|^{2q}$ by $|y|^{2q+2l}$ satisfies $0 \leq d_f \leq \Delta_f$, with that sign.

3 The four scalar tails on the original line

Consider the four positive integrals with weights

$$h_{\chi,0} = |\chi(c+iy)|^2, \quad h_{\chi,f} = |f(c+iy)\chi(c+iy)|^2, \quad h_{\text{pow},0} = |y|^{2q}, \quad h_{\text{pow},l} = |y|^{2q+2l}.$$

For each weight let $H = \int h d\sigma = H^I + H^B + H^F$ and

$$t_H = \log \left(1 + \frac{H^I + H^F}{H^B} \right) \geq 0, \quad \log H = \log H^B + t_H. \quad (\text{LET10})$$

Here every $H^B > 0$. We prove a common bound at scalar rank one rather than applying a high-rank statement outside its specified domain.

The original contour and reflection calculation BRD23–24 proves

$$C_L e^{-\pi|y|} \leq \sigma(y) \leq C_U e^{-|y|}, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}, \quad C_U = \frac{\Gamma(1/4)^2}{2\pi\sqrt{\cos 1}}, \quad C_U/C_L = (\cos 1)^{-1/2}. \quad (\text{LET11})$$

Those proofs retain the Euler contour rotation and keyhole residue; they are included in the complete source dependencies. Pairing the roots of χ as $\zeta, -\zeta$ gives

$$|\chi(c+iy)|^2 \leq (y^2 + k^2(d+g))^q.$$

On $1 \leq z \leq 2$, the reverse triangle inequality instead gives $|\chi(c+iqz)|^2 \geq q^{2q}(3/4)^q$, and $|f(c+iqz)|^2 \geq q^{2l}$. On $|z| \leq \epsilon$, LET4 gives the respective upper bounds $q^{2q}(2\epsilon^2)^q$ and $q^{2l}(2\epsilon^2)^l$. The coefficient polynomial is exactly the constant one. In particular its scalar value is bounded by $8^{2q} \int_1^2 1 dz$, with no missing coefficient directions. Integrating over the inner segment and dividing by the positive bulk subsegment gives for both actual weights

$$H^I/H^B \leq \kappa_{I,1} := \frac{2\epsilon}{\sqrt{\cos 1}} \left(\frac{4}{3} 2^{-13} e^{2\pi} \right)^q. \quad (\text{LET12})$$

Indeed the numerator/denominator ratio before enlarging it is $2\epsilon(\cos 1)^{-1/2}(4/3)^q e^{2\pi q}(2\epsilon^2)^{q+s}$, $s = 0, l$. Multiplication by $8^{2q} \geq 1$ gives LET12 because $2\epsilon^2 8^2 = 2^{-13}$ and $(2\epsilon^2)^s \leq 1$. For the reference powers the corresponding stronger constant is $2\epsilon(\cos 1)^{-1/2}\epsilon^{2s}(2^{-14}e^{2\pi})^q$, hence is at most LET12.

For the far estimate retain $D_s = 2q + 2s \leq 5q$. At $z \geq 64$, integration of $\log(z/64) \leq (z-64)/64$ gives

$$\int_{64}^{\infty} z^{D_s} e^{-qz} dz \leq \frac{64^{D_s} e^{-64q}}{q - D_s/64}, \quad q - D_s/64 \geq 59q/64. \quad (\text{LET13})$$

For the actual weights, LET7 on the bulk subsegment gives the lower factor $e^{-\eta}$, where $\eta = -a_k + \delta_\chi(1)$, and $\delta_\chi(x) = q\tau_x^2/[2(1-\tau_x)]$. The far upper weights have the extra factors $e^{\delta_\chi(64)+\Delta_{f,s}(64)}$, where $\Delta_{f,0} = 0$, $\Delta_{f,l}(x) = S_2/(q^2 x^2)$. Using LET11, both real tails, and the Jacobian $dy = q dz$, their ratio is at most

$$\frac{2}{\sqrt{\cos 1}} \frac{e^{2\pi q + \eta + \delta_\chi(64) + \Delta_{f,s}(64)} 64^{D_s} e^{-64q}}{q - D_s/64}. \quad (\text{LET14})$$

No polynomial-growth factor is needed for the constant coefficient. The original domain implies $\tau_1 \leq 1/4$ and $k \geq 5$. It gives separately

$$-a_k \leq 7q/60, \quad \delta_\chi(1) \leq q/24, \quad \delta_\chi(64) \leq q/24, \quad \Delta_{f,l}(64) \leq q/8192.$$

The last follows from $S_2 \leq 4l^3$ and $2l \leq q$. Their sum is less than $q/4$. Using $64 - 2\pi - 5 \log 64 > 20$, LET13–14 therefore give

$$H^F/H^B \leq \frac{128}{59q\sqrt{\cos 1}} e^{-19q} \leq \kappa_F := \frac{512q}{59\sqrt{\cos 1}} e^{-19q}. \quad (\text{LET15})$$

The elementary displayed strict exponent bound follows from $\pi < 4$ and $\log 2 < 1$. For the reference powers all three correction factors in LET14 equal one, so the same inequality follows directly. Thus the four actual scalar corrections obey

$$0 \leq t_H \leq L_1 := \log(1 + \kappa_{I,1} + \kappa_F). \quad (\text{LET16})$$

The constant $\kappa_{I,1}$ decays exponentially: as proved by the positive series for $\log 2$, $\log 2 > 2/3$; also $\log(4/3) < 1/3$ and $22/7 - \pi = \int_0^1 x^4(1-x)^4/(1+x^2) dx > 0$. Their substitution gives $\log((4/3)2^{-13}e^{2\pi}) < -43/21 < -2$.

The complete exact low-endpoint transfer is now

$$\log \mathfrak{r}_\chi - \log \frac{m_{2q+2l}}{m_{2q}} = d_\chi + d_f + t_{H_{\chi,f}} - t_{H_{\chi,0}} - t_{H_{\text{pow},l}} + t_{H_{\text{pow},0}}. \quad (\text{LET17})$$

It follows by writing each of the four full integrals as its own bulk integral times e^{t_H} . In particular, with $a_0 = E_0 + 2L_1$ and $b_0 = a_0 + \Delta_f$,

$$\boxed{-a_0 \leq \log \mathfrak{r}_\chi - \log \frac{m_{2q+2l}}{m_{2q}} \leq b_0.} \quad (\text{LET18})$$

The exact one-sided numerator correction occurs only in the upper bound. Since

$$|a_k| \leq |d - g|/(3e), \quad \delta_\chi \leq \frac{2(d+g)^2}{3e^3 k^2 \epsilon^4}, \quad \Delta_f \leq \frac{1}{16e^2 k \epsilon^2},$$

the finite errors in LET18 are $O(1)$ at fixed $m = 1, \delta, \gamma$ and $O(k^{-1})$ for each fixed $m \geq 2, \delta, \gamma$. The last displayed bound uses $S_2 \leq k^3/16$ and $q \geq ek^2$. Both tail corrections have the already proved exponential decay.

4 The whole original signed Gamma functional

Retain the full high reference matrices

$$(\mathbf{M}_{a,r})_{ij} = m_{2a+i+j}, \quad \mathcal{H}_{q,l,r} = \det \mathbf{M}_{q+l,r} / \det \mathbf{M}_{q,r}, \quad r = q, q+1.$$

Let

$$P_k = \log D_{l,q-1} + \log D_{l,q} - \log D_{l,2q-1} - \log D_{l,2q}, \quad D_{l,N} = \prod_{a=0}^N \prod_{j=0}^{2l-1} (a+j+1/2),$$

$$W_k = P_k - \log \frac{m_{2q+2l}}{m_{2q}} + \log \mathcal{H}_{q,l,q} + \log \mathcal{H}_{q,l,q+1}. \quad (\text{LET19})$$

All factors of the original density are retained in the moment matrices; CTR53 proves their exact rational moment calculation with mass factors displayed before cancellation. The exact old identity is $\mathfrak{T}_k = P_k - \log \mathfrak{r}_\chi + \log \mathfrak{R}_{2q-1} + \log \mathfrak{R}_{2q}$.

Write the proved high-endpoint constants from CTR45–48 as

$$A = 2E_{\chi,l} + 2(L_q + L_{q+1}), \quad B = A + (2q+1)\Delta_f, \quad E_{\chi,l} = lE_0.$$

Their complete signed sum has error in $[-A, B]$. The minus sign of the low endpoint reverses LET18. Therefore

$$\boxed{W_k - A - b_0 \leq \mathfrak{T}_k \leq W_k + B + a_0,} \quad \boxed{-W_k - B - a_0 \leq R_k^0 - R_k^\sigma \leq -W_k + A + b_0.} \quad (\text{LET20})$$

This is a direct substitution of exact determinant identities and finite proved errors. In particular the entire Gamma reference W_k depends only on the original integers q, l and the same fixed Gamma density. The original δ, γ remain in its error constants, finite threshold, original quotient, and all exact relation maps. Both sides have error $O(k) = o(q)$; at every fixed $m \geq 2$ the error is $O(1)$. No value is assigned here to the remaining explicit W_k .

Finally, preserve the exact earlier arithmetic notation $\Delta_\Gamma = \delta_\sigma - \mathfrak{T}_k$ and $B_{\text{ar}} = B_0 + \Delta_\Gamma$, with $-a_- \leq \delta_\sigma \leq a_+$. Adding these signed intervals gives the completed receiving formula

$$B_0 - W_k - B - a_0 - a_- \leq B_{\text{ar}} \leq B_0 - W_k + A + b_0 + a_+. \quad (\text{LET21})$$

The baseline B_0 in LET21 is the existing arithmetic quantity, whereas the lowercase a_0, b_0 are the low-source errors explicitly defined above. Subtracting the original $4q \log(D_h k)$ gives the corresponding interval for the unchanged HC residual.

Parity, positive shifts and the complete universal Gamma center

5 Original moments and the two sheets of the squared coordinate

Retain $k = 4l + 1, l \geq 1, m \geq 1, e = 1 + k(m - 1), q = e(k + 1)^2, c = k/2$, and the original measure and its mass

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad c_\sigma = \int_{\mathbb{R}} d\sigma = \sqrt{2\pi}. \quad (\text{PHT1})$$

Use distinct names for the ordinary and even moments:

$$m_j = \int_{\mathbb{R}} y^j d\sigma(y), \quad \mu_j = m_{2j}, \quad M_{a,r} = (m_{2a+i+j})_{0 \leq i,j < r}, \quad H_n^{(a)} = \det(\mu_{a+i+j})_{0 \leq i,j < n}. \quad (\text{PHT2})$$

Here a, n are nonnegative integers, $r \geq 1$, and $H_0^{(a)} = 1$. All moments are finite: rotation of Euler's Gamma integral through any angle $0 < \theta < \pi/2$ gives

$$\sigma(y) \leq \frac{\Gamma(1/4)^2}{2\pi\sqrt{\cos\theta}} e^{-\theta|y|}.$$

For the rotation use the argument $\text{sgn}(y)\theta$; the factor contributed by the power is $e^{-\theta|y|/2}$, and the absolute remaining Euler integral is $\Gamma(1/4)(\cos\theta)^{-1/4}$ before squaring. The small and large circular arcs vanish by the positive real exponent and by $\cos\theta > 0$. This proves the displayed bound and its constants directly.

The positive-axis and negative-axis maps to $t = y^2$ have inverse branches $y = \sqrt{t}$ and $y = -\sqrt{t}$. Each branch pushes the original measure to

$$d\nu_+(t) = d\nu_-(t) = \frac{\sigma(\sqrt{t})}{2\sqrt{t}} dt, \quad t > 0.$$

The reversed orientation of the negative branch changes the integration limits as well as its Jacobian, so its measure is positive with the same factor $1/2$. Thus the full pushforward is

$$d\nu(t) = d\nu_+(t) + d\nu_-(t) = \frac{\sigma(\sqrt{t})}{\sqrt{t}} dt, \quad \int_0^\infty d\nu = c_\sigma, \quad \mu_j = \int_0^\infty t^j d\nu(t). \quad (\text{PHT3})$$

No half-axis contribution has been treated as the whole mass.

Every polynomial has a unique decomposition $P(y) = P_e(y^2) + yP_o(y^2)$. The exact coefficient map $(P_e, P_o) \mapsto P_e(y^2) + yP_o(y^2)$ has inverse

$$P_e(t) = \frac{P(\sqrt{t}) + P(-\sqrt{t})}{2}, \quad P_o(t) = \frac{P(\sqrt{t}) - P(-\sqrt{t})}{2\sqrt{t}}. \quad (\text{PHT4})$$

Adding the two branch integrals cancels their cross terms and gives

$$\int_{\mathbb{R}} |P(y)|^2 y^{2a} d\sigma(y) = \int_0^\infty |P_e(t)|^2 t^a d\nu(t) + \int_0^\infty |P_o(t)|^2 t^{a+1} d\nu(t). \quad (\text{PHT5})$$

These same formulas extend to a unitary map of the corresponding L^2 spaces: the inverse in PHT4 is defined for almost every $t > 0$, and PHT5 proves both its range and norm. The point $t = 0$ has zero measure. On the exact multiplication domains this map transports the original real coordinate operator to

$$M_y \longleftrightarrow \begin{pmatrix} 0 & M_t \\ I & 0 \end{pmatrix}, \quad M_y^2 \longleftrightarrow \begin{pmatrix} M_t & 0 \\ 0 & M_t \end{pmatrix}. \quad (\text{PHT6})$$

Indeed multiplication of $P_e(y^2) + yP_o(y^2)$ by y gives $(tP_o)(y^2) + yP_e(y^2)$. Thus the two parity blocks retain their exact coupling under the original coordinate. Explicitly, in the base Hilbert space

$$\mathcal{K}_a = L^2(t^a d\nu) \oplus L^2(t^{a+1} d\nu),$$

the first operator has domain

$$\{(u, v) \in \mathcal{K}_a : u \in L^2(t^{a+1} d\nu), \quad v \in L^2(t^{a+2} d\nu)\}$$

and acts by $(u, v) \mapsto (tv, u)$. Its squared operator has domain

$$\{(u, v) \in \mathcal{K}_a : u \in L^2(t^{a+2} d\nu), \quad v \in L^2(t^{a+3} d\nu)\}$$

and acts by $(u, v) \mapsto (tu, tv)$. These are exactly the conditions that the displayed images belong to $\mathcal{K}_a$. For the squared domain the intermediate domain conditions follow from $t \leq 1 + t^2$. PHT5 then identifies these domains with $\{P : yP \in L^2(y^{2a} d\sigma)\}$ and $\{P : y^2 P \in L^2(y^{2a} d\sigma)\}$, respectively, including the membership of P in that base space.

6 The complete finite parity congruence

Set $n_e = \lceil r/2 \rceil, n_o = \lfloor r/2 \rfloor$. Let Π_r have columns $e_0, e_2, \dots, e_{2n_e-2}$ followed by $e_1, e_3, \dots, e_{2n_o-1}$ in the original coefficient frame. Then $\Pi_r^{-1} = \Pi_r^T$ and $\det \Pi_r = (-1)^{n_e(n_e-1)/2}$: the j -th even index crosses exactly j smaller odd indices. Evenness of σ makes each odd moment zero. Entrywise substitution therefore proves

$$\Pi_r^T M_{a,r} \Pi_r = \begin{pmatrix} (\mu_{a+i+j})_{i,j < n_e} & 0 \\ 0 & (\mu_{a+1+i+j})_{i,j < n_o} \end{pmatrix}. \quad (\text{PHT7})$$

This is a congruence in the original coefficient space. Its two determinant signs cancel, so with $D_{a,r} = \det M_{a,r}$,

$$D_{a,2n} = H_n^{(a)} H_{n+1}^{(a+1)}, \quad D_{a,2n+1} = H_{n+1}^{(a)} H_n^{(a+1)}. \quad (\text{PHT8})$$

Every factor is strictly positive: its quadratic form is the integral of a nonzero polynomial's squared modulus against a positive density on $(0, \infty)$.

The original S frame is also retained. The map $P(S) \mapsto P(c + iy)$ has coefficient matrix

$$E_{j,n} = \binom{n}{j} c^{n-j} (\sqrt{-1})^j \quad (j \leq n), \quad E^{-1} : Q(y) \mapsto Q((S - c)/i), \quad \det E = i^{r(r-1)/2}. \quad (\text{PHT9})$$

Entries below the displayed triangular range are zero. The source Gram in the S frame is exactly $E^* M_{a,r} E$, and its determinant is $D_{a,r}$ because $|\det E| = 1$. Multiplication by $(S - c)^a$ contributes the exact scalar i^a under this map; its squared modulus is one. These identities identify the moments in PHT2 with the actual monomial Gamma relation subspaces, including their degree.

7 The two original high degrees reduce to one positive shift

The stipulated $q = e(k + 1)^2$ is even; put $n = q/2$. Define

$$Z_a = H_n^{(a)} H_{n+1}^{(a)} (H_n^{(a+1)})^2, \quad A_n^{(a)} = H_n^{(a+1)} / H_n^{(a)} > 0. \quad (\text{PHT10})$$

At the two required ranks $r = q, q + 1$, PHT8 proves the exact combination

$$\frac{D_{q+l,q}}{D_{q,q}} \frac{D_{q+l,q+1}}{D_{q,q+1}} = \frac{Z_{q+l}}{Z_q}, \quad \frac{Z_{a+1}}{Z_a} = A_n^{(a)} A_{n+1}^{(a)} (A_n^{(a+1)})^2. \quad (\text{PHT11})$$

Hence its logarithm is exactly the finite sum

$$\log \frac{Z_{q+l}}{Z_q} = \sum_{j=0}^{l-1} \{ \log A_n^{(q+j)} + \log A_{n+1}^{(q+j)} + 2 \log A_n^{(q+j+1)} \}. \quad (\text{PHT12})$$

This is a contiguous-shift expression for the original two high determinants, with their unchanged ranks. The original mass factor in each $H_n^{(a)}$ is c_σ^n , so Z_a contains $c_\sigma^{4n+1} = c_\sigma^{2q+1}$. It cancels only after the numerator and denominator in PHT11 are formed.

8 Original Toda time and the squared-coordinate time

For $u \geq 0$, define the actual pushed-forward family

$$d\nu_{a,u}(t) = t^a e^{-ut} d\nu(t), \quad H_n^{(a)}(u) = \det \left(\int_0^\infty t^{a+i+j} e^{-ut} d\nu(t) \right)_{i,j < n}. \quad (\text{PHT13})$$

This is precisely the original y -source multiplied by e^{-uy^2} , pushed forward through its two branches. All moment derivatives at $u = 0$ exist from the right by dominated convergence using the original higher moments. For $\text{Re } u > 0$ they are holomorphic, by the same domination on compact subsets. The real family with $u < 0$ does not have finite mass: Gamma reflection and the Euler bound give $\sigma(y) \geq C e^{-\pi|y|}$ with $C = \pi/\Gamma(3/4)^2 > 0$, and $e^{|u|t - \pi\sqrt{t}}$ grows exponentially for sufficiently large t . This proves the actual time domain rather than assigning an unjustified two-sided strip to the quadratic flow. The reflection formula here follows on $0 < \text{Re } z < 1$ by changing $x = tv$ in the Euler double integral, obtaining $\Gamma(z)\Gamma(1-z) = \int_0^\infty t^{z-1}/(1+t) dt$. A keyhole contour with argument in $(0, 2\pi)$ has residue $e^{i\pi(z-1)}$ at -1 and straight-segment factor $1 - e^{2\pi iz}$; its vanishing circular arcs give the value $\pi/\sin(\pi z)$. At $z = 1/4 + iy/2$, take squared moduli and use $|\Gamma(3/4 - iy/2)| \leq \Gamma(3/4)$ from Euler's integral. Since $|\sin(\pi/4 + i\pi y/2)|^2 = \cosh(\pi y)/2$, this proves the stated lower bound.

Let $p_j^{(a,u)}$ be the monic orthogonal polynomials in this measure, with norms

$$h_j = H_{j+1}^{(a)}(u)/H_j^{(a)}(u).$$

Their coefficients are obtained by inverting the positive finite moment Gram, so the moment differentiability just proved also justifies each coefficient derivative below. Expansion of tp_j and orthogonality give

$$tp_j = p_{j+1} + b_j p_j + \alpha_j p_{j-1}, \quad \alpha_j = h_j/h_{j-1} > 0 \ (j \geq 1), \quad \alpha_0 = 0. \quad (\text{PHT14})$$

All other lower coefficients vanish because $\langle tp_j, p_i \rangle = \langle p_j, tp_i \rangle = 0$ for $i < j-1$; comparison against p_{j-1} gives α_j . Differentiating orthogonality under the negative tilt proves $\partial_u p_j = \alpha_j p_{j-1}$, since its coefficient against every p_i is that of tp_j for $i < j$. Also $\partial_u h_j = -b_j h_j$. Differentiating $b_j = h_j^{-1} \int tp_j^2 d\nu_{a,u}$ and using PHT14 gives

$$\partial_u b_j = \alpha_j - \alpha_{j+1}, \quad \partial_u \alpha_j = \alpha_j (b_{j-1} - b_j). \quad (\text{PHT15})$$

In detail the derivative of the numerator of b_j is $2\alpha_j h_j - \|tp_j\|^2 = h_j(\alpha_j - \alpha_{j+1} - b_j^2)$, and differentiating h_j^{-1} supplies the missing b_j^2 . Since $H_n = \prod_{j < n} h_j$, telescoping PHT15 proves

$$(\log H_n^{(a)}(u))'' = \frac{H_{n+1}^{(a)}(u)H_{n-1}^{(a)}(u)}{H_n^{(a)}(u)^2} > 0 \quad (n \geq 1), \quad (\text{PHT16})$$

or equivalently $H_n H_n'' - (H_n')^2 = H_{n+1} H_{n-1}$. At $u = 0$ these derivatives are the proved right derivatives. Thus the coefficient and sign of the Hankel-Toda equation are exactly those of the existing original-source Toda calculation.

There is an explicit map to that calculation's original linear tilt as well. For real θ , on $L^2(y^{2a}d\sigma)$, multiplication by $e^{\theta y/2}$ is the isometry from the source weighted by $e^{\theta y}$ to the original source, with inverse given by the negative exponential. Under PHT4–5 the same map is the following exact operator matrix on the two branches:

$$\begin{pmatrix} \cosh(\theta\sqrt{t}/2) & \sqrt{t} \sinh(\theta\sqrt{t}/2) \\ t^{-1/2} \sinh(\theta\sqrt{t}/2) & \cosh(\theta\sqrt{t}/2) \end{pmatrix}. \quad (\text{PHT17})$$

Substituting $y = \sqrt{t}$ and $y = -\sqrt{t}$ proves each entry and its inverse at $-\theta$. To specify the weighted source and its full cross Gram, write this matrix as $B_{\theta/2}(t)$. The two branch integrals give the pulled-back form

$$\int_0^\infty (\bar{u} \ \bar{v}) G_\theta(t) \begin{pmatrix} u \\ v \end{pmatrix} t^a d\nu(t), \quad G_\theta(t) = \begin{pmatrix} \cosh(\theta\sqrt{t}) & \sqrt{t} \sinh(\theta\sqrt{t}) \\ \sqrt{t} \sinh(\theta\sqrt{t}) & t \cosh(\theta\sqrt{t}) \end{pmatrix}. \quad (\text{PHT17a})$$

For real θ , direct matrix multiplication, using $\cosh^2 x + \sinh^2 x = \cosh(2x)$ and $2 \sinh x \cosh x = \sinh(2x)$, proves

$$B_{\theta/2}(t)^* \text{diag}(1, t) B_{\theta/2}(t) = G_\theta(t), \quad B_{-\theta/2}(t) B_{\theta/2}(t) = I.$$

Thus the space of measurable pairs with finite PHT17a norm is mapped isometrically and onto $\mathcal{K}_a$ by $B_{\theta/2}$, with inverse $B_{-\theta/2}$; the equality of forms proves the domain and range even when these matrices are unbounded on the base space. If viewed as an operator on $\mathcal{K}_a$ itself, its domain consists precisely of the pairs in $\mathcal{K}_a$ whose PHT17a integral is finite. The branch map in PHT4 identifies that source with the full original space $L^2(e^{\theta y} y^{2a} d\sigma)$. The quadratic tilt in PHT13 instead gives the diagonal multiplication $e^{-ut/2}$ in both components. Both are the specified functions of the operator in PHT6. This proves the relation between the two times on the original spaces and keeps the parity mixing of the original linear time.

For real $|\theta| < \pi/2$ the tilted y -moments and their derivatives converge: in the earlier Gamma bound choose $|\theta| < \theta_0 < \pi/2$. The same orthogonal-polynomial derivative calculation, with real variable y and positive tilt $e^{\theta y}$, applies to the exactly defined matrix $M_{a,r}(\theta)_{ij} = \int_{\mathbb{R}} y^{2a+i+j} e^{\theta y} d\sigma(y)$ and proves

$$(\log \det M_{a,r}(\theta))''|_{\theta=0} = h_r^{y,a} / h_{r-1}^{y,a}. \quad (\text{PHT18})$$

Here the superscript y, a denotes the full original real-line monic norms. Parity and PHT5 identify them exactly as $h_{2j}^{y,a} = H_{j+1}^{(a)} / H_j^{(a)}$ and $h_{2j+1}^{y,a} = H_{j+1}^{(a+1)} / H_j^{(a+1)}$. Consequently the two original Toda curvatures have the explicit positive expressions

$$\begin{aligned} \frac{h_{2n}^{y,a}}{h_{2n-1}^{y,a}} &= \frac{H_{n+1}^{(a)} H_{n-1}^{(a+1)}}{H_n^{(a)} H_n^{(a+1)}}, \\ \frac{h_{2n+1}^{y,a}}{h_{2n}^{y,a}} &= \frac{H_{n+1}^{(a+1)} H_n^{(a)}}{H_n^{(a+1)} H_{n+1}^{(a)}}. \end{aligned} \quad (\text{PHT19})$$

These are exact specializations of the original norm-ratio Toda controls at the ranks needed by the programme. The contiguous identities and their positive computation follow in the attached complete derivation.

9 Exact contiguous Gamma determinants and positive Christoffel evolution

HCT1. Original measure and the typed square map. Throughout this section

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \gamma_0 = \sqrt{2\pi}, \quad y \in \mathbb{R}.$$

Let $s : \mathbb{R} \rightarrow [0, \infty)$ be $s(y) = y^2$ and let $\nu = s_*\sigma$. For every nonnegative Borel function g , change of variable on each half of the original real line gives

$$\int_{\mathbb{R}} g(y^2) d\sigma(y) = \int_0^\infty g(t) \frac{|\Gamma(1/4 + i\sqrt{t}/2)|^2}{2\pi\sqrt{t}} dt. \quad (\text{HCT1})$$

Thus the factor $t^{-1/2}$ includes both preimages of $t > 0$. For integers $a, n \geq 0$ put

$$d\nu_a(t) = t^a d\nu(t), \quad \mu_n = \int_{\mathbb{R}} y^{2n} d\sigma(y) = \int_0^\infty t^n d\nu(t), \quad H_n^{(a)} = \det[\mu_{a+i+j}]_{i,j=0}^{n-1}, \quad H_0^{(a)} = 1.$$

No factor γ_0 is removed from any $H_n^{(a)}$.

All moments exist. Indeed, rotation by $\theta \operatorname{sgn}(y)$ in Euler's Gamma integral, with $0 < \theta < \pi/2$, gives

$$|\Gamma(1/4 + iy/2)| \leq \Gamma(1/4)(\cos \theta)^{-1/4} e^{-\theta|y|/2}.$$

The rotation is justified on a truncated sector: the small circular arc is $O(\epsilon^{1/4})$, and the large arc tends to zero because $\cos \theta > 0$. Squaring proves exponential decay of σ . The Gamma function is continuous and nonzero at $1/4$, so the density is positive on an interval about zero. Every nonzero polynomial has positive squared integral against every ν_a ; consequently

$$H_n^{(a)} > 0 \quad (a \geq 0, n \geq 0). \quad (\text{HCT2})$$

HCT2. Gamma mass, moments, and the full real-line polynomial norms. The constants needed below can be derived directly from the original Gamma weight. For $\lambda = 1/4$ set $g(x) = e^{\lambda x - e^x}$. With the Fourier convention $\widehat{g}(\xi) = \int_{\mathbb{R}} g(x) e^{i\xi x} dx$, Euler's integral gives $\widehat{g}(\xi) = \Gamma(\lambda + i\xi)$. The function g and all its derivatives decay faster than every inverse power at both ends. Fourier inversion of its autocorrelation therefore gives

$$\int_{\mathbb{R}} e^{iu\xi} |\Gamma(\lambda + i\xi)|^2 d\xi = 2\pi \int_{\mathbb{R}} g(x + u/2) g(x - u/2) dx = \frac{2\pi\Gamma(2\lambda)}{(2 \cosh(u/2))^{2\lambda}}.$$

The last equality follows by $v = e^x$ in the integral on the right. Substituting $y = 2\xi$ and $u = 2w$ gives the exact characteristic function

$$\int_{\mathbb{R}} e^{iwy} d\sigma(y) = \gamma_0 (\cosh w)^{-1/2} \quad (w \in \mathbb{R}).$$

The exponential bound in HCT1 permits its analytic continuation to the open strip $|\operatorname{Im} w| < \pi/2$. In particular

$$\int_{\mathbb{R}} e^{zy} d\sigma(y) = \gamma_0 (\cos z)^{-1/2} \quad (|z| < \pi/2), \quad \mu_n = \gamma_0 (2n)! [z^{2n}] (\cos z)^{-1/2}. \quad (\text{HCT3})$$

The square root in (HCT3) is positive at $z = 0$.

Define monic polynomials $p_j(y)$ by the convergent generating function on $|z| < 1$, with the branches analytic at zero and equal to their real values there,

$$\sum_{j=0}^{\infty} p_j(y) \frac{z^j}{j!} = (1 + z^2)^{-1/4} \exp(y \arctan z). \quad (\text{HCT4})$$

For sufficiently small real z, w , (HCT3) and

$$\cos(\arctan z + \arctan w) = \frac{1 - zw}{\sqrt{1 + z^2} \sqrt{1 + w^2}}$$

show that the integral of the product of the two generating functions is $\gamma_0(1 - zw)^{-1/2}$. Derivatives may pass under the integral by the exponential bound in HCT1. Comparison of coefficients proves

$$\int_{\mathbb{R}} p_i(y) p_j(y) d\sigma(y) = \mathbf{1}_{i=j} \gamma_0 j! (1/2)_j. \quad (\text{HCT5})$$

Here $(b)_j = b(b+1) \cdots (b+j-1)$ and $(b)_0 = 1$. Differentiating (HCT4) gives, with $p_{-1} = 0$,

$$y p_j(y) = p_{j+1}(y) + j(j-1/2) p_{j-1}(y), \quad p_0 = 1. \quad (\text{HCT6})$$

In particular p_{2n} is even and p_{2n+1} is odd. This also follows directly from $p_j(-y) = (-1)^j p_j(y)$ in (HCT4).

HCT3. Monic polynomials on the actual square-pushforward. Let $P_n^{(a)}(t)$ be the monic polynomial orthogonal to $1, t, \dots, t^{n-1}$ in $L^2(\nu_a)$, and write $h_n^{(a)} = \int (P_n^{(a)})^2 d\nu_a$. Positivity (HCT2) gives existence and uniqueness by finite Gram elimination. The determinant formula and its constant term are

$$P_n^{(a)}(t) = \frac{1}{H_n^{(a)}} \det \begin{bmatrix} \mu_a & \mu_{a+1} & \cdots & \mu_{a+n} \\ \mu_{a+1} & \mu_{a+2} & \cdots & \mu_{a+n+1} \\ \vdots & \vdots & & \vdots \\ \mu_{a+n-1} & \mu_{a+n} & \cdots & \mu_{a+2n-1} \\ 1 & t & \cdots & t^n \end{bmatrix}, \quad P_n^{(a)}(0) = (-1)^n \frac{H_n^{(a+1)}}{H_n^{(a)}}. \quad (\text{HCT7})$$

For $n = 0$ this means $P_0^{(a)} = 1$. The last-row expansion in the first column proves the displayed sign $(-1)^n$. Orthogonality follows by integrating against t^j , which duplicates a row for $0 \leq j < n$. A monic triangular change from monomials to the $P_j^{(a)}$ has determinant one, so

$$H_n^{(a)} = \prod_{j=0}^{n-1} h_j^{(a)}, \quad h_n^{(a)} = \frac{H_{n+1}^{(a)}}{H_n^{(a)}} > 0. \quad (\text{HCT8})$$

Define

$$c_n^{(a)} = \frac{H_{n+1}^{(a+1)} H_n^{(a)}}{H_{n+1}^{(a)} H_n^{(a+1)}} > 0 \quad (n \geq 0), \quad (\text{HCT9})$$

$$d_n^{(a)} = \frac{H_{n+1}^{(a)} H_{n-1}^{(a+1)}}{H_n^{(a)} H_n^{(a+1)}} > 0 \quad (n \geq 1), \quad d_0^{(a)} = 0. \quad (\text{HCT10})$$

The exact Christoffel and inverse connection identities are

$$t P_n^{(a+1)}(t) = P_{n+1}^{(a)}(t) + c_n^{(a)} P_n^{(a)}(t), \quad (\text{HCT11})$$

$$P_n^{(a)}(t) = P_n^{(a+1)}(t) + d_n^{(a)} P_{n-1}^{(a+1)}(t) \quad (n \geq 1). \quad (\text{HCT12})$$

For $n = 0$ the second identity reads $1 = 1$. To prove (HCT11), (HCT7) gives $P_{n+1}^{(a)}(0) + c_n^{(a)} P_n^{(a)}(0) = 0$. Division by t is consequently polynomial division with zero remainder. The quotient is monic of degree n and has zero ν_{a+1} inner product with t^j for $j < n$; uniqueness proves the identity. Taking its ν_a inner product with $P_n^{(a+1)}$ proves

$$h_n^{(a+1)} = c_n^{(a)} h_n^{(a)}. \quad (\text{HCT13})$$

Indeed the $P_{n+1}^{(a)}$ term has zero inner product, while $P_n^{(a+1)} - P_n^{(a)}$ has degree at most $n-1$. For (HCT12), $P_n^{(a)} - P_n^{(a+1)}$ has degree at most $n-1$. Its ν_{a+1} inner products with all polynomials of degree at most $n-2$ vanish, because multiplication of such a polynomial by t has degree at most $n-1$. It is therefore a multiple of $P_{n-1}^{(a+1)}$. The numerator of that coefficient is

$$\int P_n^{(a)} P_{n-1}^{(a+1)} d\nu_{a+1} = \int P_n^{(a)} t P_{n-1}^{(a+1)} d\nu_a = h_n^{(a)},$$

using (HCT11) at $n-1$. Dividing by $h_{n-1}^{(a+1)}$ gives exactly (HCT10), including its positive sign.

HCT4. Positive discrete Toda identities. Combining (HCT11) and (HCT12) twice, entirely within the original polynomial spaces, gives

$$tP_n^{(a)} = P_{n+1}^{(a)} + (c_n^{(a)} + d_n^{(a)})P_n^{(a)} + d_n^{(a)}c_{n-1}^{(a)}P_{n-1}^{(a)}. \quad (\text{HCT14})$$

At $n = 0$ the last term is absent. Expanding instead in the $a + 1$ basis gives

$$tP_n^{(a+1)} = P_{n+1}^{(a+1)} + (c_n^{(a)} + d_{n+1}^{(a)})P_n^{(a+1)} + c_n^{(a)}d_n^{(a)}P_{n-1}^{(a+1)}.$$

Coefficient comparison with (HCT14) at $a + 1$ proves

$$c_n^{(a+1)} + d_n^{(a+1)} = c_n^{(a)} + d_{n+1}^{(a)} \quad (n \geq 0), \quad (\text{HCT15})$$

$$c_{n-1}^{(a+1)}d_n^{(a+1)} = c_n^{(a)}d_n^{(a)} \quad (n \geq 1). \quad (\text{HCT16})$$

All nonzero factors in these identities are positive. In the notation $b_n^{(a)} = c_n^{(a)} + d_n^{(a)}$ and $u_n^{(a)} = d_n^{(a)}c_{n-1}^{(a)}$, the two recurrences give the complete b, u coefficients and their shifted values, including $b_n^{(a+1)} = c_n^{(a)} + d_{n+1}^{(a)}$ and $u_n^{(a+1)} = c_n^{(a)}d_n^{(a)}$.

Here is the exact polynomial meaning of the matrix factorization. Use column vectors

$$\mathbf{P}^{(a)} = (P_0^{(a)}, P_1^{(a)}, \dots)^T$$

as formal polynomial sequences. Let L_a have diagonal 1 and subdiagonal $(L_a)_{n,n-1} = d_n^{(a)}$, and let U_a have diagonal $c_n^{(a)}$ and superdiagonal 1. Every row has finite support. Then (HCT11)–(HCT12) state

$$\mathbf{P}^{(a)} = L_a \mathbf{P}^{(a+1)}, \quad t\mathbf{P}^{(a+1)} = U_a \mathbf{P}^{(a)}, \quad J_a = L_a U_a, \quad J_{a+1} = U_a L_a. \quad (\text{HCT17})$$

Consequently $J_{a+1}U_a = U_a J_a$ and $J_a L_a = L_a J_{a+1}$. These are row-finite polynomial identities; no infinite-dimensional bounded-operator assertion or truncation of an endpoint row is needed. At a finite degree the identities retain every displayed P_{n+1} term. The scalar multiplication map is literally $t : \mathbb{R}[t]_{\leq n} \rightarrow \mathbb{R}[t]_{\leq n+1}$.

HCT5. Exact initial Gamma data and a subtraction-free finite evolution. Parity, (HCT1), and uniqueness give the exact source identifications

$$p_{2n}(y) = P_n^{(0)}(y^2), \quad p_{2n+1}(y) = yP_n^{(1)}(y^2). \quad (\text{HCT18})$$

Thus (HCT5) gives, with the original mass retained,

$$h_n^{(0)} = \gamma_0(2n)!(1/2)_{2n}, \quad H_n^{(0)} = \gamma_0^n \prod_{j=0}^{n-1} (2j)!(1/2)_{2j}, \quad (\text{HCT19})$$

$$h_n^{(1)} = \gamma_0(2n+1)!(1/2)_{2n+1}, \quad H_n^{(1)} = \gamma_0^n \prod_{j=0}^{n-1} (2j+1)!(1/2)_{2j+1}. \quad (\text{HCT20})$$

Equations (HCT9)–(HCT10) therefore yield the exact positive rational starting values

$$c_n^{(0)} = (2n+1)(2n+1/2), \quad d_n^{(0)} = 2n(2n-1/2), \quad n \geq 0. \quad (\text{HCT21})$$

For a fixed step $a \rightarrow a + 1$ introduce auxiliary numbers e_n by

$$e_0 = c_0^{(a)}, \quad d_0^{(a+1)} = 0, \quad c_0^{(a+1)} = e_0 + d_1^{(a)}, \quad (\text{HCT22})$$

$$e_n = \frac{c_n^{(a)}e_{n-1}}{e_{n-1} + d_n^{(a)}}, \quad d_n^{(a+1)} = \frac{c_n^{(a)}d_n^{(a)}}{e_{n-1} + d_n^{(a)}}, \quad c_n^{(a+1)} = e_n + d_{n+1}^{(a)} \quad (n \geq 1). \quad (\text{HCT23})$$

Every denominator and every $e_n, c_n^{(a+1)}$ is strictly positive; $d_n^{(a+1)} > 0$ for $n \geq 1$. These formulas use only positive sums, products, and quotients. To verify them, (HCT15) at zero gives (HCT22). If $c_{n-1}^{(a+1)} = e_{n-1} + d_n^{(a)}$, then (HCT16) gives the middle formula in (HCT23). The first formula implies $e_n + d_n^{(a+1)} = c_n^{(a)}$; substituting this in (HCT15) gives the last formula. This proves the recurrence and the claimed positivity by induction.

It follows, without determinant cancellation, that for any finite integer $a, n \geq 0$

$$H_n^{(a)} = H_n^{(0)} \prod_{s=0}^{a-1} \prod_{j=0}^{n-1} c_j^{(s)}. \quad (\text{HCT24})$$

Indeed $H_n^{(a+1)} / H_n^{(a)} = \prod_{j < n} c_j^{(a)}$ follows from (HCT8) and (HCT13), and then telescopes in a . An initial array through index $N + A$ suffices to compute all $c_n^{(a)}, d_n^{(a)}$ with $a \leq A, n \leq N$: one may retain indices $0 \leq n \leq N + A - a$ after step a and shrink the retained interval by one at each subsequent step. Thus the formulas give a finite exact calculation for arbitrarily specified original ranks and shifts. The assertion here concerns an exact finite recurrence, with no unproved assertion about its computational cost or large-rank limit.

HCT6. Sign-correct contiguous determinants. For $n \geq 1$, apply the Desnanot–Jacobi identity to the $(n+1) \times (n+1)$ matrix $[\mu_{a+i+j}]_{i,j=0}^n$, deleting its first and last rows and columns. The two principal minors are $H_n^{(a+2)}$ and $H_n^{(a)}$, the two cross minors are both $H_n^{(a+1)}$, and the central minor is $H_{n-1}^{(a+2)}$. Their original index order is retained. Hence

$$H_n^{(a)} H_n^{(a+2)} - (H_n^{(a+1)})^2 = H_{n+1}^{(a)} H_{n-1}^{(a+2)} > 0. \quad (\text{HCT25})$$

For completeness, the determinant identity used here follows by Schur-complement elimination of the central block, whose determinant is positive by (HCT2). The remaining 2×2 determinant is the product of its diagonal entries minus the product of its off-diagonal entries; multiplication by the central determinant squared gives (HCT25). For $n = 1$ the central block has size zero and determinant one, giving the same identity directly.

Put $\kappa_0^{(a)} = 0$ and, for $n \geq 1$,

$$\kappa_n^{(a)} = \frac{H_{n+1}^{(a)} H_{n-1}^{(a+2)}}{(H_n^{(a+1)})^2} > 0. \quad (\text{HCT26})$$

Then the exact logarithmic second difference is

$$\log H_n^{(a+2)} - 2 \log H_n^{(a+1)} + \log H_n^{(a)} = \log(1 + \kappa_n^{(a)}). \quad (\text{HCT27})$$

For $n \geq 1$ its equivalent positive Toda expression is

$$0 < \frac{d_n^{(a)}}{c_{n-1}^{(a+1)}} = \frac{\kappa_n^{(a)}}{1 + \kappa_n^{(a)}} < 1, \quad \log(1 + \kappa_n^{(a)}) = -\log\left(1 - \frac{d_n^{(a)}}{c_{n-1}^{(a+1)}}\right). \quad (\text{HCT28})$$

Substitution of (HCT9)–(HCT10), followed by (HCT25), proves every equality and inequality in (HCT28). In particular, if $R_n^{(a)} = H_n^{(a+1)} / H_n^{(a)}$, then

$$R_n^{(a)} = \prod_{j=0}^{n-1} c_j^{(a)}, \quad \frac{R_n^{(a+1)}}{R_n^{(a)}} = 1 + \kappa_n^{(a)}. \quad (\text{HCT29})$$

For a finite integer $l \geq 1$, exact telescoping gives

$$\log \frac{H_n^{(a+l+1)} / H_n^{(a+1)}}{H_n^{(a+l)} / H_n^{(a)}} = \sum_{s=0}^{l-1} \log(1 + \kappa_n^{(a+s)}). \quad (\text{HCT30})$$

This quantity is strictly positive when $n \geq 1$. The signed claim in (HCT27)–(HCT30) concerns a -differences of the indicated determinant ratios. The degree difference is the different exact identity $\log R_{n+1}^{(a)} - \log R_n^{(a)} = \log c_n^{(a)}$; no sign or asymptotic size of this last logarithm is needed.

HCT7. Return to both original full matrices. Retain the full matrices from the original real coordinate:

$$M_{a,r}[i,j] = \int_{\mathbb{R}} y^{2a+i+j} d\sigma(y) \quad (0 \leq i, j < r), \quad \mathcal{D}_{a,r} = \det M_{a,r}. \quad (\text{HCT31})$$

Let $n_0 = \lceil r/2 \rceil$, $n_1 = \lfloor r/2 \rfloor$. Permute the original indices to $0, 2, 4, \dots, 1, 3, 5, \dots$ using the same permutation in the source and its dual. Odd moments vanish by evenness of σ . The even-even block is $[\mu_{a+i+j}]_{i,j < n_0}$ and the odd-odd block is $[\mu_{a+1+i+j}]_{i,j < n_1}$. Thus the exact congruence and its determinant are

$$P_r M_{a,r} P_r^T = \text{diag}([\mu_{a+i+j}]_{i,j < n_0}, [\mu_{a+1+i+j}]_{i,j < n_1}), \quad \mathcal{D}_{a,r} = H_{n_0}^{(a)} H_{n_1}^{(a+1)}. \quad (\text{HCT32})$$

The determinant sign is $\det(P_r)^2 = 1$. At the function-space level the same map sends the coefficient vector of $p(t) + yq(t)$, with $t = y^2$, to the two coefficient vectors of (p, q) . Its exact norm identity is

$$\int_{\mathbb{R}} y^{2a} |p(y^2) + yq(y^2)|^2 d\sigma(y) = \int_0^\infty |p(t)|^2 d\nu_a(t) + \int_0^\infty |q(t)|^2 d\nu_{a+1}(t).$$

The cross term is odd. This proves the typed orthogonal direct-sum map with both original parity sectors retained.

For the even integer $q = e(k+1)^2 = 2n$ specified by the packet, the two original ranks are therefore exactly

$$\mathcal{D}_{a,q} = H_n^{(a)} H_n^{(a+1)}, \quad \mathcal{D}_{a,q+1} = H_{n+1}^{(a)} H_n^{(a+1)}. \quad (\text{HCT33})$$

Only the stipulated integrality and evenness of q are used here; the packet parameter e is left unchanged. In particular

$$\frac{\mathcal{D}_{a,q+1}}{\mathcal{D}_{a,q}} = h_n^{(a)}, \quad \frac{\mathcal{D}_{a+l,q+1}/\mathcal{D}_{a,q+1}}{\mathcal{D}_{a+l,q}/\mathcal{D}_{a,q}} = \prod_{s=0}^{l-1} c_n^{(a+s)}. \quad (\text{HCT34})$$

The empty product at $l = 0$ is one.

Write $Z_a = \mathcal{D}_{a,q} \mathcal{D}_{a,q+1}$ while keeping the two original ranks fixed. Equations (HCT29), (HCT33) give

$$Z_a = H_n^{(a)} H_{n+1}^{(a)} (H_n^{(a+1)})^2, \quad (\text{HCT35})$$

$$\frac{Z_{a+1}}{Z_a} = R_n^{(a)} R_{n+1}^{(a)} (R_n^{(a+1)})^2, \quad (\text{HCT36})$$

$$\frac{Z_{q+l}}{Z_q} = \prod_{s=0}^{l-1} \left\{ \left(\prod_{j=0}^{n-1} c_j^{(q+s)} \right) \left(\prod_{j=0}^n c_j^{(q+s)} \right) \left(\prod_{j=0}^{n-1} c_j^{(q+s+1)} \right)^2 \right\}. \quad (\text{HCT37})$$

Equation (HCT37) evaluates the paired determinant ratio at the requested original exponents $a = q, q + l$ by the positive rational evolution (HCT21)–(HCT23). Each of the four factors supplied by the two full matrices remains visible. The γ_0 factors in (HCT19)–(HCT24) cancel in this particular ratio because both numerator and denominator have the same total matrix dimension; they are present in the individual determinants throughout.

Finally the exact curvature of this same paired determinant is

$$\frac{Z_{a+2} Z_a}{Z_{a+1}^2} = (1 + \kappa_n^{(a)})(1 + \kappa_{n+1}^{(a)})(1 + \kappa_n^{(a+1)})^2 > 1. \quad (\text{HCT38})$$

For $l \geq 1$ the consequent finite shift identity is

$$\log \frac{Z_{a+l+1}/Z_{a+1}}{Z_{a+l}/Z_a} = \sum_{s=0}^{l-1} [\log(1 + \kappa_n^{(a+s)}) + \log(1 + \kappa_{n+1}^{(a+s)})]$$

$$+ 2 \log(1 + \kappa_n^{(a+s+1)})] > 0. \quad (\text{HCT39})$$

These statements follow by adding (HCT27) for exactly the four blocks in (HCT35), exponentiating for (HCT38), and telescoping for (HCT39); they make no replacement of q by an independent large parameter. They provide the full finite return, its positive contiguous increments, and its exact degree-step comparison. A growth rate along the packet values of q, k, l requires evaluating these explicit positive quantities; no such rate is assumed in any identity above.

10 The complete universal Gamma center is a positive rational product

Use exactly the all-index moments m_j in PHT2. The universal center identified in the low-endpoint transport calculation is the following finite quantity:

$$\begin{aligned} \mathcal{D}_{l,N} &= \prod_{a=0}^N \prod_{j=0}^{2l-1} (a + j + 1/2), \\ P_k &= \log \mathcal{D}_{l,q-1} + \log \mathcal{D}_{l,q} - \log \mathcal{D}_{l,2q-1} - \log \mathcal{D}_{l,2q}, \\ W_k &= P_k - \log \frac{m_{2q+2l}}{m_{2q}} + \log \frac{D_{q+l,q}}{D_{q,q}} + \log \frac{D_{q+l,q+1}}{D_{q,q+1}}. \end{aligned} \quad (\text{PHT20})$$

Here $\mathcal{D}_{l,N}$ denotes exactly the source product called $D_{l,N}$ in GCR and LET, while $D_{a,r}$ retains its moment-determinant definition from PHT8. Define

$$G_{q,l} = \left(\prod_{a=q}^{2q-1} \prod_{j=0}^{2l-1} (a + j + 1/2) \right) \left(\prod_{a=q+1}^{2q} \prod_{j=0}^{2l-1} (a + j + 1/2) \right) > 0. \quad (\text{PHT21})$$

Cancellation of the explicitly overlapping monic products gives $e^{P_k} = G_{q,l}^{-1}$. The low moment ratio retains its precise index conversion:

$$\frac{m_{2q+2l}}{m_{2q}} = \frac{\mu_{q+l}}{\mu_q} = \frac{H_1^{(q+l)}}{H_1^{(q)}} = \prod_{s=0}^{l-1} c_0^{(q+s)}. \quad (\text{PHT22})$$

The last equality is HCT13 at degree zero and then finite telescoping. Combining PHT11, HCT37 and PHT22 proves

$$\begin{aligned} W_k &= \log \mathcal{W}_{q,l}, \\ \mathcal{W}_{q,l} &= \frac{1}{G_{q,l}} \prod_{s=0}^{l-1} \frac{\left(\prod_{j=0}^{n-1} c_j^{(q+s)} \right) \left(\prod_{j=0}^n c_j^{(q+s)} \right) \left(\prod_{j=0}^{n-1} c_j^{(q+s+1)} \right)^2}{c_0^{(q+s)}}, \quad n = q/2. \end{aligned} \quad (\text{PHT23})$$

Every factor is positive and rational: the initial c, d values are the exact rational Gamma values HCT21, and HCT22–23 use only positive sums, products and quotients. The initial norms still contain c_σ , as HCT19–20 record. Their common powers cancel in PHT11 and PHT22 only after their full source determinants are constructed.

Thus PHT23 is a finite evaluation procedure for the entire universal center in PHT20, requiring neither a new moment quadrature nor a difference of nearly equal Gram determinants. One sufficient finite initial range is $0 \leq j \leq n + q + l$: HCT22–23 then compute all needed coefficients through shift $q + l$ with the proved decreasing-index range in HCT24. This statement specifies an exact finite procedure. It does not assert a bit-complexity estimate or an asymptotic value along the original coupled sequence $q = e(k+1)^2, l = (k-1)/4$. The positive contiguous identities HCT25–39 provide the exact equations for that next calculation, with all ranks and original source factors retained.